Relevant Research Papers:

Module 1: Automated compliance checking (the closest thing to our core idea)

1. **"NLP-based Automated Compliance Checking of Data Processing Agreements against GDPR"**

   Amaral, Azeem, Abualhaija, Briand — *IEEE Transactions on Software Engineering, 2023*
   arxiv.org/abs/2209.09722

   This paper extracts "shall" requirements from GDPR provisions relevant to Data Processing Agreement compliance and uses NLP to check documents against them automatically, in close collaboration with legal experts. This is the closest existing work to what you're building. Your differentiation: they check documents against GDPR, you check *code and APIs* against the DPDP Act. Cite this as your primary related work.

   Link - [2209.09722] NLP-based Automated Compliance Checking of Data Processing Agreements against GDPR

2. **"Democratizing GDPR Compliance: AI-Driven Privacy Policy Interpretation"**
   ACM International Conference on Contemporary Computing, 2024
   dl.acm.org/doi/10.1145/3675888.3676142

   This paper directly references the NLP-based automated compliance checking approach as a foundation for AI-driven interpretation of privacy policies against GDPR obligations. Cite this to show your work sits in an active 2024 research stream.

   Link - **ACM Other conferences**

3. **"Enhancing GDPR Compliance Through NLP: Automated Policy Evaluation and Legal Document Analysis"**

   Okonicha, Sadovykh — *Springer Lecture Notes in Networks and Systems, 2026*
   link.springer.com/chapter/10.1007/978-3-031-94770-4_34

   This very recent work combines NLP-based policy evaluation with GDPR legal document analysis, using approaches like sentence classification and semantic similarity to identify compliance gaps in privacy policies. Cite this as the most recent prior work. Then say: no equivalent exists for India's DPDP Act — that's your gap.

   Link - **Springer**

Module 2: Privacy Policy Analysis with LLM

4. **"Using LLMs for Automated Privacy Policy Analysis: Prompt Engineering, Fine-Tuning and Explainability"** ***arXiv, 2025*** **arxiv.org/pdf/2503.16516**

   This paper is among the first to systematically evaluate LLM-based classifiers for automated privacy policy analysis, finding that combining prompt engineering and fine-tuning consistently outperforms prior NLP methods across multiple privacy policy corpora and taxonomies — and that there is very little prior research on using LLMs for this purpose. This directly validates your LLM layer design choice. Cite it when your professor asks why you used an LLM rather than a traditional classifier**.**

   Link - **arxiv**

5. **"AI-enabled Automation for Completeness Checking of Privacy Policies" Amaral, Abualhaija, Torre, Sabetzadeh, Briand —** ***IEEE Transactions on Software Engineering, 2021***

   This paper proposes AI-enabled automation for completeness checking of privacy policies, establishing a model-based approach for verifying whether privacy policies satisfy regulatory requirements — an approach cited extensively in subsequent compliance research. This is a foundational paper your lit review must include**.**

   Link - **arxiv**

6. **"Have You Been Properly Notified? Automatic Compliance Analysis of Privacy Policy Text with GDPR Article 13" Liu et al. —** ***Proceedings of The Web Conference, 2021***
   Referenced in: researchgate.net/publication/387458781

   Checks whether a privacy policy properly discloses what Article 13 requires. The direct conceptual ancestor of your consent gap detector — they check policy vs GDPR article, you check policy vs actual system behaviour.

Module 3 — Consent gap detection (most novel module)

7. **"Breaking the Illusion: Automated Reasoning of GDPR Consent Violations"**
   arXiv, 2025
   arxiv.org/pdf/2512.22789

   This paper explicitly studies the consistency between data usage *practices* and data usage *statements* such as privacy policies, covering data collection, sharing, and withdrawal, but notes that existing works focus on standalone consent mechanisms like cookie banners and overlook how consent is solicited in functional systems. This confirms your consent gap detection is a recognised open problem. Quote this gap directly in your introduction.

   Link - **arxiv**

8. **"Between Law and Code: Challenges and Opportunities for Automating Privacy Assessments"**
   Automated Software Engineering, Springer, 2026
   link.springer.com/article/10.1007/s10515-026-00601-4

   This paper confirms that while automated privacy analysis tools demonstrate feasibility, a research gap remains in how such automation can bridge the distance between legal requirements and actual technical implementations, noting that privacy assessors face significant challenges when existing tools don't connect law to code. This is a strong citation for your motivation section. "Law to code" is exactly what your DPDP mapper and consent gap detector do.

   Link - **Springer**

9. **"Data Protection by Design Tool for Automated GDPR Compliance Verification Based on Semantically Modeled Informed Consent"**
   MDPI Sensors, 2022
   mdpi.com/1424-8220/22/7/2763

   This paper presents a scalable tool for automated compliance verification based on informed consent modeled as a knowledge graph, implementing a regulation-to-code process that translates GDPR regulations into software-verifiable measures, and showing how automated compliance checks can trigger notification alerts on detected consent violations. The "regulation-to-code" phrase is useful language for your paper. Their approach uses knowledge graphs; yours uses LLM reasoning, cite as contrast**.**

   Link - **MDPI**

Module 4 — PII detection and classification

10. **"A Hybrid Rule-Based NLP and Machine Learning Approach for PII Detection and Anonymization in Financial Documents"**
    Scientific Reports, Nature, 2025
    nature.com/articles/s41598-025-04971-9

    This paper proposes a scalable hybrid approach combining rule-based NLP, machine learning, and a custom Named Entity Recognition model for PII detection and anonymization, achieving a precision of 94.7% and F1-score of 91.1% on financial documents. Directly applicable to your PII classifier design. Cite the hybrid NER + rules approach as the established method you're building on.

    Link - **Nature**

11. **"Privacy-Preserving Personal Identifiable Information (PII) Label Detection Using Machine Learning"** ***IEEE Conference, 2023***
    researchgate.net/publication/375880997

    This paper proposes a method for PII tag detection that combines machine learning with regular expressions, noting that PII identification is particularly difficult due to the different types, forms, and inconsistent definitions across various data protection laws. The inconsistency across laws point is a good justification for why you need a DPDP-specific classifier rather than a generic one.

    Link - **ResearchGate**

Module 5 — LLM for legal compliance reasoning

12. **"LegiLM: A Fine-Tuned Legal Language Model for Data Compliance"**
    arXiv, 2024
    arxiv.org/html/2409.13721v1

    LegiLM is fine-tuned on specialized legal corpora and GDPR-specific datasets, and through rigorous benchmarking has demonstrated effectiveness in interpreting complex legal documents and assessing GDPR compliance, showing that domain-specific training significantly improves over general-purpose LLMs for legal tasks. This validates using an LLM for compliance reasoning. Your approach uses a general LLM with prompt engineering rather than fine-tuning, you can frame this as a practical tradeoff for a capstone scope.

    Link - **arXiv**

13. **"GraphCompliance: Aligning Policy and Context Graphs for LLM-Based Regulatory Compliance"**
    arXiv, 2025
    arxiv.org/pdf/2510.26309

    This paper introduces PrivComp-KG, which leverages knowledge graphs combined with LLMs for compliance verification against privacy regulations — showing that structured regulatory knowledge combined with LLM reasoning outperforms either approach alone. Cite to position your LangChain + structured DPDP mapper design as aligned with the state of the art.

    Link - **arxiv**

14. **"Rethinking Legal Compliance Automation: Opportunities with Large Language Models"**
    Hassani, Sabetzadeh, Amyot, Liao — *IEEE International Requirements Engineering Conference (RE), 2024*
    Referenced in: arxiv.org/pdf/2510.26309

    A 2024 IEEE RE paper specifically making the case for LLMs in legal compliance automation. Cite this when your professor questions whether LLMs are appropriate for a legal domain.

15. **DPDP Act — background and motivation**

    "Reviewing the Privacy Implications of India's Digital Personal Data Protection Act (2023)"
    ResearchGate, 2024
    researchgate.net/publication/377768426

    This paper identifies challenges organizations face in adhering to India's DPDP Act alongside global regulations, highlighting that the rapid expansion of data-driven systems has created an urgent need for automated compliance tools, and that modern AI and machine learning technologies are well-positioned to address

regulatory compliance at scale. Your primary citation for establishing the DPDP Act as a real and unmet compliance challenge.

Link - ResearchGate

16. **"The Digital Personal Data Protection Act of 2023: Strengthening Privacy in the Digital Age"**
Vijayalakshmi — *IJLCW, 2024*
researchgate.net/publication/387550185

This paper notes that India's Data Protection Board is empowered to levy penalties of up to ₹250 crore for violations of the DPDP Act, establishing the significant financial stakes that make compliance tooling urgent for Indian companies. Use this to open your motivation section. ₹250 crore per breach is a compelling number.

Link - ResearchGate

Layer 1 — General automated compliance (Amaral et al. 2023, Okonicha et al. 2026) — shows this is a real research area with published IEEE/ACM work.

Layer 2 — Privacy policy NLP (Liu et al. 2021, LLM paper 2025) — shows the specific technique of checking policies with ML is established.

Layer 3 — Consent gap and code-level checking (Breaking the Illusion 2025, Between Law and Code 2026) — shows your core contribution addresses a confirmed research gap.

Layer 4 — India DPDP context (Vijayalakshmi 2024, Reviewing Privacy Implications 2024) — shows no automated tool exists for this specific jurisdiction.

research gap statement: *"Existing automated compliance tools focus on GDPR policy-text analysis. No prior work addresses automated, code-aware compliance checking against India's DPDP Act 2023, nor the automated detection of gaps between a company's privacy notice and its actual technical implementation."*